The exact (S_1, S_2) range of pure symmetric five-qubit states

A compact rigorous proof

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Abstract

For pure symmetric five-qubit states we determine the complete attainable set of the one- and two-body sector lengths (S_1, S_2). The lower boundary consists of a short algebraic arc, a square-root arc, and the line $S_2 = 2S_1$; the upper boundary is $S_2 = 10$. The only substantial reduction is to prove that a fixed-S_1 minimizer can be chosen with collective C_2 symmetry. We give the finite spin-5/2 certificates needed for that reduction, solve the remaining one-variable problem exactly, and use connectedness of the moment-map fibers to fill the region.

1 Notation and statement

Let

$$|\psi\rangle = \sum_{k=0}^{5} d_k |D_5^{(k)}\rangle, \qquad \sum_{k=0}^{5} |d_k|^2 = 1,$$

be a pure symmetric five-qubit state. For a Pauli string $\sigma_{\boldsymbol{\mu}}$, define

$$S_r(\psi) = \sum_{\mathrm{wt}(\boldsymbol{\mu})=r} |\langle\psi|\sigma_{\boldsymbol{\mu}}|\psi\rangle|^2.$$

We identify the symmetric subspace with spin $j = 5/2$, so that

$$J_z|D_5^{(k)}\rangle = \left(\frac{5}{2} - k\right)|D_5^{(k)}\rangle, \qquad J_+|D_5^{(k)}\rangle = \sqrt{k(6-k)}\,|D_5^{(k-1)}\rangle.$$

The spin, Dicke, and Majorana descriptions used below are reviewed in Refs. [2, 3]. Put

$$s_a = \langle J_a\rangle, \qquad M_{ab} = \left\langle \frac{J_aJ_b + J_bJ_a}{2} \right\rangle, \qquad Q = M - \frac{35}{12} I_3,$$

and write $m = |s|$ and $\mathcal{Q} = \operatorname{tr} Q^2$.

Lemma 1.1 (Moment and multipole dictionary)**.** *For every pure symmetric five-qubit state,*

$$\boxed{S_1 = \frac{4}{5}m^2, \qquad S_2 = \frac{10}{3} + \frac{2}{5}\mathcal{Q}.} \tag{1}$$

With the Hilbert–Schmidt orthonormal convention

$$T_{LM}^{(5)} = \sqrt{\frac{2L+1}{6}} \sum_{m,m'} C^{\frac{5}{2}m'}_{\frac{5}{2}m,LM} |\tfrac{5}{2}, m\rangle\langle\tfrac{5}{2}, m'|,$$

and, for $\widehat{\rho} = |\psi\rangle\langle\psi|$,

$$\widehat{\rho} = \sum_{L=0}^{5}\sum_{M=-L}^{L} \rho_{LM} T_{LM}^{(5)}, \qquad \rho_{LM} = \operatorname{tr}\left(\widehat{\rho}\, T_{LM}^{(5)\dagger}\right), \qquad a_L = \sum_{M=-L}^{L} |\rho_{LM}|^2,$$

one has

$$\boxed{a_0 = \frac{1}{6}, \qquad a_1 = \frac{S_1}{14} = \frac{2}{35}|s|^2, \qquad a_2 = \frac{5}{112}\left(S_2 - \frac{10}{3}\right) = \frac{1}{56}\operatorname{tr} Q^2.} \tag{2}$$

Thus $S_1 = 14a_1$ and $S_2 = 20a_0 + \frac{112}{5}a_2$; moreover $\sum_{L=0}^{5} a_L = \operatorname{tr}(\widehat{\rho}^2) = 1$.

Proof. Permutation symmetry gives $\langle \sigma_a^{(i)} \rangle = 2s_a/5$, hence

$$S_1 = 5 \sum_a (2s_a/5)^2 = \frac{4}{5}|s|^2.$$

If $t_{ab} = \langle \sigma_a^{(1)} \sigma_b^{(2)} \rangle$, then

$$M = \frac{5}{4} I_3 + 5t, \qquad t = \frac{1}{3} I_3 + \frac{1}{5} Q.$$

There are ten pairs of qubits, so

$$S_2 = 10 \operatorname{tr}(t^2) = 10 \left(\frac{1}{3} + \frac{1}{25} \operatorname{tr} Q^2 \right).$$

For the tensor normalization above,

$$\rho_{10} = \sqrt{\frac{2}{35}} s_z, \qquad \rho_{1,\pm 1} = \mp \frac{s_x \pm \mathrm{i} s_y}{\sqrt{35}},$$

$$\rho_{20} = \frac{3Q_{zz}}{4\sqrt{21}}, \qquad \rho_{2,\pm 1} = \mp \frac{Q_{xz} \pm \mathrm{i} Q_{yz}}{2\sqrt{14}}, \qquad \rho_{2,\pm 2} = \frac{Q_{xx} - Q_{yy} \pm 2\mathrm{i} Q_{xy}}{4\sqrt{14}}.$$

Using $\operatorname{tr} Q = 0$ and summing absolute squares gives $a_1 = 2|s|^2/35$ and

$$a_2 = |\rho_{20}|^2 + 2|\rho_{21}|^2 + 2|\rho_{22}|^2 = \frac{1}{56} \operatorname{tr} Q^2.$$

Finally $T_{00}^{(5)} = I_6/\sqrt{6}$, so $a_0 = 1/6$. □

Set

$$p_c = 2 - \frac{3}{\sqrt{5}}, \qquad p_0 = \frac{\sqrt{5}}{3}, \qquad U_c = \frac{120\sqrt{5} - 205}{109}, \qquad x_c = \frac{U_c^2}{5} = \frac{22805 - 9840\sqrt{5}}{11881}. \tag{3}$$

For $p_c \le p \le p_0$, define

$$u(p) = -\frac{15\sqrt{5}p^4 + 20p^3 + 18\sqrt{5}p^2 - 28p - 9\sqrt{5}}{3\sqrt{5}p^4 + 12p^3 + 18\sqrt{5}p^2 - 20p + 3\sqrt{5}}, \tag{4}$$

$$\rho(p) = \frac{2p(5p^2 + 6\sqrt{5}p - 11)}{3\sqrt{5}p^2 + 2p + 3\sqrt{5}}, \tag{5}$$

$$\Phi(p, r) = \frac{2(25p^4 + 10p^2 r + 10p^2 - 24\sqrt{5}pr + 19r^2 + 58r + 9)}{5(1 + p^2 + r)^2}. \tag{6}$$

On $p_c \le p \le p_0$, one has $\rho(p) \ge 0$, with equality only at $p = p_c$.

Theorem 1.2 (Complete range)**.** *The set of pairs (S_1, S_2) attained by pure symmetric five-qubit states is*

$$\boxed{\mathcal{S}_5^{\mathrm{sym}} = \{(x, y) : 0 \le x \le 5,\ b(x) \le y \le 10\},} \tag{7}$$

where

$$\boxed{b(x) = \begin{cases} \Phi\big(p, \rho(p)\big), & 0 \le x \le x_c, \quad x = u(p)^2/5, \\ \dfrac{3x - 2\sqrt{5x} + 15}{4}, & x_c \le x \le 9/5, \\ 2x, & 9/5 \le x \le 5. \end{cases}} \tag{8}$$

The parameter $p \in [p_c, p_0]$ is unique in the first line. The three lower pieces are attained respectively by

$$d_{v_2}(p) = \frac{(-p, 0, \sqrt{\rho(p)}, 0, 1, 0)}{\sqrt{1 + p^2 + \rho(p)}}, \tag{9}$$

$$d_\infty(\lambda) = \frac{(0, 1, 0, 0, 0, \lambda)}{\sqrt{1 + \lambda^2}}, \quad \lambda = \sqrt{\frac{3 - \sqrt{5x}}{5 + \sqrt{5x}}}, \tag{10}$$

$$d(\theta) = (\cos\theta, \sin\theta, 0, 0, 0, 0), \quad x = 5 - \frac{16}{5} \sin^4\theta. \tag{11}$$

Here $p_c \le p \le p_0$, $0 \le \lambda \le p_c$, and $0 \le \theta \le \pi/2$. The upper edge is attained by

$$|\mathrm{GHZ}(5,\varphi)\rangle = \cos\frac{\varphi}{2}\, D_5^{(0)} + \sin\frac{\varphi}{2}\, D_5^{(5)}. \tag{12}$$

Here $0 \le \varphi \le \pi/2$, and $S_1 = 5\cos^2\varphi$, $S_2 = 10$, so this covers the entire upper edge. In particular, the remaining vertical boundary is $S_1 = 0$, $18/5 \le S_2 \le 10$.

At the four lower endpoints or junctions one may take

x	$b(x)$	equality state
0	18/5	$\psi_0 = \frac{1}{4}\left(\sqrt{5}\, D_5^{(0)} + \sqrt{2}\, D_5^{(2)} - 3D_5^{(4)}\right)$
x_c	$b(x_c)$	$d_{v_2}(p_c)$ or $d_\infty(p_c)$
9/5	18/5	$D_5^{(1)}$
5	10	$D_5^{(0)}$.

The states $d_{v_2}(p_0)$ and ψ_0 differ only by a global phase and a collective z-rotation.

2 Exact minimization in a Dicke-parity sector

We first minimize $\mathcal{Q}$ at fixed m under the temporary assumption of Dicke parity. Spin-5/2 time reversal acts as

$$\Theta\left(\sum_{k=0}^{5} z_k D_5^{(k)}\right) = \sum_{k=0}^{5} (-1)^k \overline{z}_k D_5^{(5-k)}.$$

It exchanges the parity sectors, sends s to $-s$, and preserves Q. It therefore suffices to consider

$$|\psi\rangle = z_0 D_5^{(0)} + z_2 D_5^{(2)} + z_4 D_5^{(4)}.$$

Parity implies $s_x = s_y = Q_{xz} = Q_{yz} = 0$. Writing $q_z = Q_{zz}$ and $\zeta = \langle J_+^2\rangle$, direct calculation gives

$$\mathcal{Q} = \frac{3}{2}q_z^2 + \frac{1}{2}|\zeta|^2, \qquad \zeta = 2\sqrt{10}\,\overline{z}_0 z_2 + 6\sqrt{2}\,\overline{z}_2 z_4. \tag{13}$$

At fixed moduli, the reverse triangle inequality minimizes $|\zeta|$ by making its two summands antiparallel. Put

$$\sigma = \langle J_z\rangle, \qquad R = \frac{3}{4} + \frac{\sigma}{2}, \qquad t = |z_2|^2.$$

Normalization and fixed σ give

$$|z_0|^2 = \frac{R-t}{2}, \qquad |z_4|^2 = \frac{2-R-t}{2}, \qquad 0 \le t \le \min(R, 2-R). \tag{14}$$

This closed interval includes all projective faces. After phase minimization,

$$\mathcal{Q} \ge \mathcal{F}_R(t), \qquad \mathcal{F}_R(t) = 6\left(R - \frac{1}{3}\right)^2 + (44 - 32R)t - 4t^2 - 12t\sqrt{5(R-t)(2-R-t)}, \tag{15}$$

and equality is attainable.

Lemma 2.1 (Strict convexity)**.** *For fixed R, $\mathcal{F}_R$ is strictly convex in the interior of its physical interval.*

Proof. Let $w = 1 - t$ and $c = |1 - R|$. In the interior $0 \le c < w \le 1$, and

$$\mathcal{F}_R''(t) = \frac{4\{3\sqrt{5}[2w^3 - 3c^2 w + c^2] - 2(w^2 - c^2)^{3/2}\}}{(w^2 - c^2)^{3/2}}.$$

For $d = c^2$, set $L = 2w^3 - 3dw + d$ and $\Delta = (w^2 - d)^{3/2}$. Then

$$\frac{\partial}{\partial d}\frac{L}{\Delta} = \frac{w^2 + \frac{1}{2}d(1-3w)}{(w^2-d)^{5/2}} \ge 0.$$

For $w \le 1/3$ this is immediate; for $w \ge 1/3$, use $d \le w^2$ to bound the numerator below by $\frac{3}{2}w^2(1-w)$. Hence $L/\Delta \ge 2$, and the numerator of $\mathcal{F}_R''$ is at least $(6\sqrt{5} - 2)\Delta > 0$. □

Set

$$U = 2m = \sqrt{5S_1}. \tag{16}$$

For the orientation $\sigma = -U/2$, put $R_-(U) = (3-U)/4$. For $0 \le U < 3$,

$$\mathcal{F}'_{R_-(U)}(0) = 20 + 8U - 3\sqrt{5(3-U)(5+U)}, \tag{17}$$

whereas the derivative tends to $+\infty$ at the other endpoint. Since the two terms compared in (17) are positive,

$$\operatorname{sgn} \mathcal{F}'_{R_-(U)}(0) = \operatorname{sgn}(109U^2 + 410U - 275).$$

The unique nonnegative zero is U_c from (3). Lemma 2.1 therefore shows that the minimizer is interior for $U < U_c$ and is $t = 0$ for $U_c \le U \le 3$.

For the interior solution, phase choices put

$$d = \frac{(-p, 0, \sqrt{r}, 0, 1, 0)}{\sqrt{1 + p^2 + r}},$$

and (14) gives

$$p^2 = \frac{R_-(U) - t}{2 - R_-(U) - t}, \qquad r = \frac{2t}{2 - R_-(U) - t} = \frac{3 - U - (5+U)p^2}{1+U}. \tag{18}$$

Direct reduction of $\mathcal{F}'_{R_-(U)}(t) = 0$ gives $U = u(p)$ and $r = \rho(p)$. Since

$$u(p_0) = 0, \qquad u(p_c) = U_c, \qquad \rho(p_c) = 0,$$

the interior minimum is exactly the first arc in (8). For $U_c \le U \le 3$, the endpoint gives

$$D(U) = \frac{15}{4} - \frac{U}{2} + \frac{3U^2}{20} = \frac{3S_1 - 2\sqrt{5S_1} + 15}{4}. \tag{19}$$

This value is admissible for every $0 \le U \le 3$, because

$$\eta_U = \frac{D_5^{(1)} + \lambda_U D_5^{(5)}}{\sqrt{1 + \lambda_U^2}}, \qquad \lambda_U = \sqrt{\frac{3-U}{5+U}}, \tag{20}$$

satisfies $S_1(\eta_U) = U^2/5$ and $S_2(\eta_U) = D(U)$.

Comparison of the two orientations

In this subsection we use the same rational expressions (4)–(5) on their natural extension $p \ge p_c$. In the affine chart set

$$v = -2\langle J_z \rangle = \frac{3 - 5p^2 - r}{1 + p^2 + r}.$$

The interior stationary curve is $v = u(p)$ and is strictly decreasing on the relevant interval. Indeed,

$$u'(p) = -\frac{8\sqrt{5}\,\mathcal{M}(p)}{3\mathcal{A}(p)^2} < 0, \tag{21}$$

where, for $\delta = p - \sqrt{5}/5 \ge 0$,

$$\begin{aligned} \mathcal{A}(p) &= \delta^4 + \frac{8\sqrt{5}}{5}\delta^3 + \frac{48}{5}\delta^2 + \frac{128\sqrt{5}}{75}\delta + \frac{16}{15} > 0, \\ \mathcal{M}(p) &= \delta^6 + \frac{24\sqrt{5}}{5}\delta^5 + \frac{84}{5}\delta^4 + \frac{512\sqrt{5}}{75}\delta^3 + \frac{368}{25}\delta^2 + \frac{896\sqrt{5}}{125}\delta + \frac{1088}{375} > 0. \end{aligned} \tag{22}$$

Moreover $p_c > \sqrt{5}/5$, $u(p_c) = U_c$, $u(p_0) = 0$, $u(1) = -1$, and $\rho(p) > 0$ for $p_0 \le p \le 1$.

For $0 < U \le U_c$, let

$$u(p_-(U)) = U, \qquad u(p_+(U)) = -U, \qquad p_c \le p_- < p_0 < p_+ < 1.$$

The probability coordinate is regular because

$$\frac{\mathrm{d}t}{\mathrm{d}p} = -\frac{p(v+1)}{(p^2-1)^2} \neq 0.$$

Thus strict convexity makes $p_\pm$ the respective global minima. If $\mathcal{G}(v)$ is the stationary value of S_2, direct differentiation gives

$$\mathcal{G}'(u(p)) = u(p)\mathcal{R}(p), \qquad \mathcal{R}(p) = \frac{4(p+\sqrt{5})(5p-3\sqrt{5})}{5(5p^2+2\sqrt{5}p+9)}, \tag{23}$$

and

$$\mathcal{R}'(p) = \frac{192(p+\sqrt{5}/5)}{(5p^2+2\sqrt{5}p+9)^2} > 0.$$

Consequently

$$H(U) := \mathcal{G}(-U) - \mathcal{G}(U), \qquad H(0) = 0, \qquad H'(U) = U\left[\mathcal{R}(p_+) - \mathcal{R}(p_-)\right] > 0.$$

The opposite orientation is strictly higher for $0 < U \le U_c$.

For $U_c \le U \le 3$, put

$$R_+(U) = \frac{3+U}{4}, \qquad q_D(U) = \frac{5}{2}\left(D(U) - \frac{10}{3}\right) = \frac{(3U-5)^2}{24}.$$

Every physical t satisfies $0 \le t \le 1$, and

$$\mathcal{F}_{R_+(U)}(t) - q_D(U) = A(U,t) - 12t\sqrt{5(R_+ - t)(2 - R_+ - t)}, \tag{24}$$

where

$$A(U,t) = \frac{1}{2}\{U(5-16t) + 8t(5-t)\} > 0.$$

For $t \le 5/16$ positivity is immediate; for $t \ge 5/16$, physicality gives $U \le 5-4t$, whence $2A \ge 56t^2 - 60t + 25 > 0$. Squaring (24) is therefore legitimate. The resulting polynomial is

$$\begin{aligned} P(U,t) &:= A(U,t)^2 - 720t^2(R_+ - t)(2 - R_+ - t) \\ &= \left(109t^2 - 40t + \frac{25}{4}\right)U^2 + (64t^3 - 430t^2 + 100t)U \\ &\qquad - 704t^4 + 1280t^3 - 275t^2. \end{aligned} \tag{25}$$

Its leading coefficient is positive and

$$\operatorname{disc}_U P = 135t^2(4t-5)(4t-1)(12t-5)^2. \tag{26}$$

Hence $P \ge 0$ for $1/4 \le t \le 1$. Possible equality points follow from

$$P(U, \tfrac{1}{4}) = \frac{(7U-1)^2}{16}, \qquad P(U, \tfrac{5}{12}) = \frac{1225(3U-5)^2}{1296}.$$

For $0 \le t \le 1/4$, put $x = 4t$ and $B_{k,n}(x) = \binom{n}{k}x^k(1-x)^{n-k}$. Exact Bernstein certificates are

$$\begin{aligned} \partial_U P(\tfrac{1}{2}, \tfrac{x}{4}) &= \sum_{k=0}^{3} d_k B_{k,3}(x), & (d_k) &= \left(\frac{25}{4}, \frac{45}{4}, \frac{153}{16}, \frac{35}{16}\right), \\ P(\tfrac{1}{2}, \tfrac{x}{4}) &= \sum_{k=0}^{7} e_k B_{k,7}(x), & (e_k) &= \left(\frac{25}{16}, \frac{335}{112}, \frac{1363}{448}, \frac{5157}{2240}, \frac{1431}{1120}, \frac{89}{224}, \frac{5}{448}, \frac{25}{64}\right). \end{aligned} \tag{27}$$

All coefficients are positive. Moreover

$$U_c - \frac{1}{2} = \frac{240\sqrt{5} - 519}{218} > 0$$

(squaring the positive sides gives $288000 > 269361$), and

$$\partial_U^2 P = 2\left(109t^2 - 40t + \frac{25}{4}\right) > 0,$$

we obtain $P > 0$ for $U \ge U_c$ and $0 \le t \le 1/4$. Thus $\mathcal{F}_{R_+}(t) \ge q_D(U)$ throughout the high interval. Equality occurs only at

$$U = \frac{5}{3}, \qquad t = \frac{5}{12}, \qquad S_1 = \frac{5}{9}, \qquad \mathcal{Q} = 0;$$

the other double root $U = 1/7$ lies below U_c.

Finally, the parity minimum is at most $18/5$, with strict inequality for $0 < U < 3$. On the first arc,

$$\frac{\mathrm{d}}{\mathrm{d}U}\mathcal{G}(U) = U\mathcal{R}(p_-(U)) < 0,$$

because $p_-(U) \le p_0 < 3\sqrt{5}/5$. On the second arc,

$$D(U) - \frac{18}{5} = \frac{(U-3)(3U-1)}{20} < 0 \qquad (U_c \le U < 3).$$

By (1), this is equivalent to $\mathcal{Q} \le 2/3$, strictly for $0 < U < 3$.

Proposition 2.2 (Parity-restricted minimum)**.** *For $0 \le S_1 \le 9/5$, the minimum over both Dicke-parity sectors and both orientations is the first two pieces of* (8)*. The two orientations tie only at $S_1 = 0$ and at the scalar point $S_1 = 5/9$.*

3 Reduction to a C_2-symmetric minimizer

We now remove the parity assumption. We use three consequences of Proposition 2.2: for every $m \le 3/2$ there is a parity state with the same m and $\mathcal{Q} \le 2/3$; the trial value $D(U)$ in (19) is admissible for every $0 \le U \le 3$; and the parity minimum is strictly below $18/5$ when $0 < U < 3$.

KKT equation and alignment

Let ψ be a global minimizer at fixed $0 < S_1 < 9/5$. At a regular point of the constraint the Euler–Lagrange equation is

$$(B - \mu\, s\cdot J)\psi = \kappa\psi, \qquad B = Q_{ab}K_{ab}, \qquad K_{ab} = \frac{1}{2}(J_aJ_b + J_bJ_a), \tag{28}$$

where $\mu = 2\lambda$ if λ is the multiplier of S_1, and $\kappa = \mathcal{Q} - \mu m^2$. At a singular point,

$$(s\cdot J - m^2)\psi = 0.$$

Since $m > 0$, such a state is a rotated Dicke state and already has parity. Hence we may assume regularity. Put

$$T_a = \frac{1}{2}\langle\{J_a, B\}\rangle.$$

Taking real J_a matrix elements in (28) gives

$$s \times T = \mu\, s \times Qs. \tag{29}$$

We now derive a second relation between these two cross products. The first step uses time reversal; the second is a finite spin-5/2 polynomial identity.

Time reversal and the Kramers plane. With the convention in Section 2, Θ is antiunitary and satisfies

$$\Theta^2 = -I, \qquad \Theta J_a \Theta^{-1} = -J_a, \qquad \langle\Theta u|v\rangle = -\langle\Theta v|u\rangle.$$

In particular $\langle\Theta\psi|\psi\rangle = 0$, so $P = |\psi\rangle\langle\psi| + |\Theta\psi\rangle\langle\Theta\psi|$ is the orthogonal projector onto a two-dimensional plane. If $C = C^\dagger$ is time-even, meaning $\Theta C\Theta^{-1} = C$, its off-diagonal matrix element on this plane vanishes:

$$\langle\Theta\psi|C\psi\rangle = -\langle\Theta(C\psi)|\psi\rangle = -\langle C\Theta\psi|\psi\rangle = -\langle\Theta\psi|C\psi\rangle = 0.$$

Its two diagonal elements agree, since antiunitarity and Hermiticity give $\langle\Theta\psi|C|\Theta\psi\rangle = \langle\psi|C|\psi\rangle$. Thus $PCP = \langle C\rangle P$. Apply this to $B = Q_{ab}K_{ab}$: each K_{ab} is time-even, and

$$\langle B\rangle = Q : M = Q : \left(Q + \frac{35}{12}I_3\right) = \mathcal{Q}, \qquad PBP = \mathcal{Q}P.$$

Here Q is held fixed at the state under consideration and $\operatorname{tr} Q = 0$ was used. Projecting (28) onto $\Theta\psi$ therefore gives

$$0 - \mu\langle\Theta\psi|s\cdot J|\psi\rangle = \kappa\langle\Theta\psi|\psi\rangle = 0.$$

Consequently, provided $\mu \neq 0$,

$$\langle\Theta\psi|s\cdot J|\psi\rangle = 0. \tag{30}$$

The spin-$5/2$ identity in coordinates. Since $m > 0$, rotate to $s = m\hat{z}$ and write

$$v = \sum_{k=0}^{5} \sqrt{\binom{5}{k}}\, c_k D_5^{(k)}, \qquad N = \langle v|v\rangle = \sum_{k=0}^{5} \binom{5}{k} |c_k|^2, \qquad \psi = \frac{v}{\sqrt{N}}.$$

Thus the c_k are weighted polynomial coefficients, not normalized Dicke amplitudes. The time-reversal convention gives

$$\langle \Theta v| = \sum_{k=0}^{5} (-1)^{5-k} \sqrt{\binom{5}{k}}\, c_{5-k} \langle D_5^{(k)}|.$$

The absence of complex conjugation on c_{5-k} is due to the antiunitarity of Θ followed by passage to the bra. In particular, pairing the terms with indices k and $5-k$ yields

$$\langle \Theta v|J_z|v\rangle = \sum_{k=0}^{5} (-1)^{5-k} \binom{5}{k} \left(\frac{5}{2} - k\right) c_{5-k} c_k = -5c_0c_5 + 15c_1c_4 - 10c_2c_3 =: \tau.$$

Equation (30) now reads $m\tau/N = 0$; since $m > 0$ and $N > 0$, it implies $\tau = 0$.

To make the remaining matrix calculation explicit, introduce

$$f = c_0c_4 - 4c_1c_3 + 3c_2^2, \qquad g = c_1c_5 - 4c_2c_4 + 3c_3^2, \qquad A_x = f + g, \quad A_y = f - g.$$

For the transition vector $w_a = \langle \Theta v|J_a|v\rangle$, the ladder matrix elements stated in Section 1 give

$$\langle \Theta v|J_+|v\rangle = -10g, \qquad \langle \Theta v|J_-|v\rangle = 10f, \qquad w = (5A_y,\, 5\mathrm{i}A_x,\, \tau).$$

The required Fierz identity, valid before imposing $s_x = s_y = 0$, is

$$N^2(T - 4Qs) = \frac{10}{3} N\widetilde{s} + 2\mathrm{i}\, w \times \overline{w}, \qquad \widetilde{s}_a = \langle v|J_a|v\rangle = Ns_a.$$

Here and below s, Q, T are evaluated in $v/\sqrt{N}$. For an explicit way to check this quartic identity, set

$$\widetilde{Q}_{ab} = \langle v|K_{ab}|v\rangle - \frac{35}{12} N\delta_{ab} = NQ_{ab}.$$

Then, with repeated indices summed over x, y, z, its left-hand side has components

$$N^2(T - 4Qs)_a = \frac{1}{2} \widetilde{Q}_{bc} \langle v|\{J_a, K_{bc}\}|v\rangle - 4\widetilde{Q}_{ab}\widetilde{s}_b.$$

Substituting the ladder matrix elements and collecting terms gives, for $a = x, y$, respectively,

$$\frac{10}{3} N\widetilde{s}_x - 10\big(A_x\overline{\tau} + \overline{A}_x\tau\big),$$
$$\frac{10}{3} N\widetilde{s}_y - 10\mathrm{i}\big(A_y\overline{\tau} - \overline{A}_y\tau\big).$$

These are also exactly the x, y components of the stated cross-product formula. Since the chosen frame has $\widetilde{s}_x = \widetilde{s}_y = 0$, we obtain the certificate

$$N^2(T - 4Qs)_x = -20\,\mathrm{Re}(A_x\overline{\tau}),$$
$$N^2(T - 4Qs)_y = 20\,\mathrm{Im}(A_y\overline{\tau}), \qquad \langle \Theta v|J_z|v\rangle = \tau. \tag{31}$$

Combining the two relations. For $\mu \neq 0$ we already proved $\tau = 0$. Equation (31) therefore gives $(T - 4Qs)_x = (T - 4Qs)_y = 0$. No assertion about the z component is needed: because $s = m\hat{z}$,

$$s \times (T - 4Qs) = m\big(-(T - 4Qs)_y,\, (T - 4Qs)_x,\, 0\big) = 0.$$

Thus $s \times T = 4s \times Qs$. Subtracting this relation from (29) yields

$$(\mu - 4)s \times Qs = 0 \qquad (\mu \neq 0).$$

The projection argument did not cover $\mu = 0$, so that case must be retained separately. Both cases can be expressed as

$$\boxed{\mu(\mu - 4)s \times Qs = 0.} \tag{32}$$

The factor μ records this separate case; it does not arise from dividing by a possibly vanishing quantity. In particular, if $\mu \notin \{0, 4\}$, then Qs is parallel to s, so the mean-spin direction is a principal axis of Q. A nonaligned regular KKT state must therefore have $\mu = 0$ or $\mu = 4$.

The exceptional multiplier $\mu = 0$

For every real traceless Q, direct evaluation in a principal frame gives

$$B^3 - 14\mathcal{Q}B - \frac{80}{3}\operatorname{tr}(Q^3)I = 0. \tag{33}$$

Indeed, if $q = Q_{zz}$ and $\delta = (Q_{xx} - Q_{yy})/2$, the even parity block of B is

$$\begin{pmatrix} 5q & \sqrt{10}\delta & 0 \\ \sqrt{10}\delta & -4q & 3\sqrt{2}\delta \\ 0 & 3\sqrt{2}\delta & -q \end{pmatrix},$$

whose characteristic polynomial is $t^3 - 14\operatorname{tr}(Q^2)t - \frac{80}{3}\operatorname{tr}(Q^3)$; the odd block has the same spectrum.

If $\mu = 0$, then $B\psi = \mathcal{Q}\psi$. Differentiate (33) in an arbitrary traceless direction and take its expectation in ψ. Since $\langle \delta B \rangle = Q : \delta Q$, one obtains

$$3\mathcal{Q}(\mathcal{Q} - 14)Q = 80\left(Q^2 - \frac{\mathcal{Q}}{3}I\right). \tag{34}$$

For $Q \neq 0$, this forces Q to be axial. Substitution of its spectrum $(a, -a/2, -a/2)$ gives

$$(3\mathcal{Q} - 2)(3\mathcal{Q} - 32)(3\mathcal{Q} - 50) = 0.$$

The three cases are respectively the Kramers doublets $|m_n| = 3/2, 1/2, 5/2$. Only the $|m_n| = 1/2$ doublet can be nonaligned. In that doublet

$$PJ_xP = \frac{3}{2}\sigma_x, \qquad PJ_yP = \frac{3}{2}\sigma_y, \qquad PJ_zP = \frac{1}{2}\sigma_z,$$

so every codeword has $m \leq 3/2$, whereas $\mathcal{Q} = 32/3$. Proposition 2.2 supplies, at the same m, a state with $\mathcal{Q} \leq 2/3$. Thus a nonaligned $\mu = 0$ state is not a global minimum.

The spin-one covariant and the multiplier $\mu = 4$

Let P_1 be the total-spin-one projector in $V_{5/2} \otimes V_{5/2}$ and $p_1 = \|P_1(\psi \otimes \psi)\|^2$. On $\operatorname{Sym}^2 V_{5/2}$, the operator $X = \sum_a J_a \otimes J_a$ has eigenvalues $-31/4, -11/4, 25/4$, hence

$$P_1 = \frac{(X + 11/4)(X - 25/4)}{70}.$$

Moreover,

$$\langle X \rangle = m^2, \qquad \langle X^2 \rangle = \sum_{a,b} \langle J_a J_b \rangle^2 = \mathcal{Q} + \frac{1225}{48} - \frac{1}{2}m^2.$$

Taking its expectation gives

$$\boxed{S_2 - 2S_1 = 28p_1, \qquad \mathcal{Q} - 4m^2 + \frac{25}{3} = 70p_1.} \tag{35}$$

Thus a $\mu = 4$ KKT state is a normalized critical point of p_1.

We record now the zero set of this covariant. Fix the Majorana convention

$$\gamma_k = \frac{(-1)^k d_k}{\sqrt{\binom{5}{k}}}, \qquad f(x, y) = \sum_{k=0}^{5} \binom{5}{k} \gamma_k x^{5-k} y^k. \tag{36}$$

Up to nonzero factors, the three components of $P_1(\psi \otimes \psi)$ are

$$C_+ = \gamma_0\gamma_4 - 4\gamma_1\gamma_3 + 3\gamma_2^2, \quad C_0 = \gamma_0\gamma_5 - 3\gamma_1\gamma_4 + 2\gamma_2\gamma_3,$$

$$C_- = \gamma_1\gamma_5 - 4\gamma_2\gamma_4 + 3\gamma_3^2.$$

Lemma 3.1 (Fourth transvectant and multiplicity)**.** *One has $p_1 = 0$ if and only if $f = L^4 M$ for linear forms L, M. More generally, if a Majorana point has multiplicity $r > 5/2$, then*

$$S_1 \geq \frac{4}{5}\left(r - \frac{5}{2}\right)^2. \tag{37}$$

Proof. The implication $f = L^4 M \Rightarrow C_+ = C_0 = C_- = 0$ follows by direct differentiation. Conversely, by the $\mathrm{SL}(2,\mathbb{C})$ covariance of the fourth transvectant, we may send any root to $x = 0$, so $\gamma_5 = 0$. If $\gamma_4 = 0$, the equations force $\gamma_2 = \gamma_3 = 0$ and $f = x^4(\gamma_0 x + 5\gamma_1 y)$. If $\gamma_4 \neq 0$, they give

$$\gamma_2 = \frac{3\gamma_3^2}{4\gamma_4}, \qquad \gamma_1 = \frac{\gamma_3^3}{2\gamma_4^2}, \qquad \gamma_0 = \frac{5\gamma_3^4}{16\gamma_4^3},$$

and substitution factors f as $Cx(Ax + By)^4$.

For the bound, rotate the repeated point to the north pole. The state then lies in $\mathrm{span}\{D_5^{(0)}, \dots, D_5^{(5-r)}\}$, whose J_z eigenvalues are at least $r - 5/2$. Hence $m \geq \langle J_z \rangle \geq r - 5/2$, and (1) proves (37). □

In particular $p_1 = 0$ implies $S_1 \geq 9/5$. It remains to show alignment when $p_1 > 0$. Let

$$\Gamma : V_{5/2} \otimes V_{5/2} \longrightarrow V_1$$

be the Clebsch–Gordan coisometry, with $\Gamma^\dagger \Gamma = P_1$, and define

$$c = \Gamma(\psi \otimes \psi), \qquad L_\psi v = \Gamma(\psi \otimes v), \qquad R = L_\psi L_\psi^\dagger, \qquad p = \|c\|^2.$$

The total-spin-one summand lies in $\mathrm{Sym}^2 V_{5/2}$, so $\Gamma(v \otimes w) = \Gamma(w \otimes v)$. Moreover, R is an equivariant linear function of $|\psi\rangle\langle\psi|$. Criticality gives $L_\psi^\dagger c = p\psi$ and $Rc = pc$. Covariance and the multiplicity-free decomposition $\mathrm{End}(V_1) = V_0 \oplus V_1 \oplus V_2$ give

$$R = \frac{1}{6} I + \frac{1}{35} s \cdot \ell - \frac{2}{35} Q : K^{(1)}. \tag{38}$$

For completeness, its coefficients are fixed by the coherent state $|5/2, 5/2\rangle$, for which

$$R = \mathrm{diag}(\tfrac{1}{7}, \tfrac{5}{14}, 0), \qquad s \cdot \ell = \mathrm{diag}(\tfrac{5}{2}, 0, -\tfrac{5}{2}), \qquad Q : K^{(1)} = \mathrm{diag}(\tfrac{5}{3}, -\tfrac{10}{3}, \tfrac{5}{3}).$$

If $p > 0$, (38) and $Rc = pc$ imply

$$\left(Q : K^{(1)} - \frac{1}{2} s \cdot \ell \right) c = \xi c, \qquad \xi \in \mathbb{R}.$$

For the normalized boost

$$\psi(t) = \frac{e^{t\, u \cdot J} \psi}{\| e^{t\, u \cdot J} \psi \|},$$

intertwining gives $\dot{c} = (u \cdot \ell - 2u \cdot s)c$; criticality gives $\langle c | \ell | c \rangle = 2ps$. In the Cartesian model write $c = x + \mathrm{i}y$ with $x \perp y$. Then $s \parallel x \times y$. Since $Q : K^{(1)} = -Q$ and $s \cdot \ell = -\mathrm{i} s \times$, real and imaginary parts give

$$-Qx - \frac{1}{2} s \times y = \xi x, \qquad -Qy + \frac{1}{2} s \times x = \xi y.$$

The cross terms are parallel to x and y, respectively, so x, y, s are eigenvectors of Q. Hence

$$p_1 > 0 \quad \Longrightarrow \quad Qs \parallel s. \tag{39}$$

Together with Lemma 3.1, this excludes a nonaligned $\mu = 4$ minimizer for $0 < S_1 < 9/5$.

From alignment to parity

Rotate an aligned state so that $s = m\hat{z}$ and diagonalize Q in the transverse plane:

$$Q = \mathrm{diag}(-q/2 + b, -q/2 - b, q), \qquad A = -\frac{3q}{2}, \qquad h = \mu m.$$

Up to an irrelevant scalar, the KKT operator is

$$H' = -AJ_z^2 + b(J_x^2 - J_y^2) - hJ_z.$$

It commutes with Dicke parity. In the ordered bases

$$(D_5^{(0)}, D_5^{(2)}, D_5^{(4)}), \qquad (D_5^{(1)}, D_5^{(3)}, D_5^{(5)}),$$

its even and odd blocks are

$$H_e = \begin{pmatrix} -25A/4 - 5h/2 & \sqrt{10}b & 0 \\ \sqrt{10}b & -A/4 - h/2 & 3\sqrt{2}b \\ 0 & 3\sqrt{2}b & -9A/4 + 3h/2 \end{pmatrix},$$

$$H_o = \begin{pmatrix} -9A/4 - 3h/2 & 3\sqrt{2}b & 0 \\ 3\sqrt{2}b & -A/4 + h/2 & \sqrt{10}b \\ 0 & \sqrt{10}b & -25A/4 + 5h/2 \end{pmatrix}.$$

Their characteristic resultant is

$$135h^3[h^2 - 16(A^2 - b^2)][h^2 - 4(A^2 - b^2)]^2. \tag{40}$$

If it is nonzero, a KKT eigenvector belongs to one parity block.

At a nonaxial crossing $h \neq 0$ and $b \neq 0$, so both blocks are irreducible real Jacobi matrices and have simple spectra. The vanishing resultant implies $A^2 > b^2$. After replacing (A, b, h, E) by its negative if necessary, take $A > 0$ and write

$$A = \frac{u^2 + v^2}{2}, \qquad b = \frac{u^2 - v^2}{2}, \qquad uv(u^2 - v^2) \neq 0.$$

By (40), the signs of u, v may be chosen so that either $h = 4uv$ or $h = 2uv$. For a common eigenvalue E, let $\epsilon_3 = (0, 0, 1)^T$ and set

$$\widetilde{e} = \mathrm{adj}(EI - H_e)\epsilon_3, \qquad \widetilde{o} = \mathrm{adj}(EI - H_o)\epsilon_3.$$

On both crossing surfaces,

$$(\widetilde{e})_1 = (\widetilde{o})_1 = \frac{3\sqrt{5}}{2}(u - v)^2(u + v)^2 \neq 0,$$

so these are nonzero eigenvectors. For $h = 4uv$, $E = 15(u^2 + v^2)/8$ and

$$\widetilde{e}^T J_x \widetilde{o} = 90\sqrt{5}\, u^4(u + v)^4, \qquad \widetilde{e}^T (J_y/\mathrm{i})\widetilde{o} = -90\sqrt{5}\, v^4(u + v)^4. \tag{41}$$

For $h = 2uv$, the common eigenvalue satisfies

$$F := -64E^2 - 16E(u^2 + v^2) + 255u^4 - 450u^2v^2 + 255v^4 = 0.$$

With

$$C_R = 8E + 25u^2 - 15v^2, \quad C_I = -8E + 15u^2 - 25v^2, \quad C_0 = \frac{9\sqrt{5}}{4}(u - v)(u + v)^3,$$

direct adjugate multiplication gives

$$\widetilde{e}^T J_x \widetilde{o} = C_0 u^2 C_R, \qquad \widetilde{e}^T (J_y/\mathrm{i})\widetilde{o} = -C_0 v^2 C_I. \tag{42}$$

Neither C_R nor C_I vanishes on the crossing, because

$$F|_{C_R=0} = -320u^2(u^2 - v^2), \qquad F|_{C_I=0} = 320v^2(u^2 - v^2).$$

Thus both transverse dipole transitions are nonzero.

Normalize the real eigenvectors and write $\psi = \alpha e + \beta o$. With $z = \overline{\alpha}\beta$,

$$s_x = 2\,\mathrm{Re}(z) e^T J_x o, \qquad s_y = -2\,\mathrm{Im}(z) e^T (J_y/\mathrm{i}) o.$$

Alignment forces $z = 0$, hence $\alpha = 0$ or $\beta = 0$. Thus every nonaxial aligned crossing state has definite parity.

It remains to inspect $h = 0$ or $b = 0$. Since $m > 0$, $h = 0$ implies $\mu = 0$. The nonaligned $|m_n| = 1/2$ codewords are nonoptimal by the preceding comparison, while the aligned codewords in that doublet already have parity about their mean-spin axis. A genuinely mixed aligned $h = 0$ eigenspace can therefore only come from the $|m_n| = 3/2$ or $|m_n| = 5/2$ doublets, or from $Q = 0$. The first two have their mean spin along the quadrupole axis and occur among the $b = 0$ branches 14 and 05 below.

If $b = 0$, away from the scalar case a degenerate eigenspace contains at most two Dicke levels. Gap one violates $s_x = s_y = 0$; the remaining odd gaps give, up to time reversal,

$$03, \quad 25, \quad 14, \quad 05,$$

with

$$S_2^{03/25} = 5 \mp 2U + \frac{3U^2}{5}, \qquad S_2^{14} = \frac{18}{5}, \qquad S_2^{05} = 10. \tag{43}$$

The smaller first value satisfies

$$5 - 2U + \frac{3U^2}{5} - D(U) = \frac{(3U-5)^2}{20} \geq 0.$$

Since $D(U)$ is an admissible parity value, this branch is not lower than the parity minimum; equality occurs only at $U = 5/3$. The 14 branch is strictly higher for $0 < U < 3$, and 05 is the upper edge. Only the scalar case $Q = 0$ remains.

Lemma 3.2 (Scalar replacement)**.** *If a pure spin-5/2 state has $Q = 0$, then $|s| = 5/6$ and, after a rotation, it has the same first and second moments as*

$$\chi = \sqrt{\frac{5}{6}}\, D_5^{(1)} + e^{\mathrm{i}\theta}\sqrt{\frac{1}{6}}\, D_5^{(5)}. \tag{44}$$

Proof. For the Kramers projector P, time reversal allows only tensor ranks $0, 2, 4$. Since $Q = 0$ removes rank two, $P = I/3 + A^{(4)}$. The identity $P^2 = P$ gives

$$\|A^{(4)}\|^2 = \frac{4}{3}, \qquad (A^{(4)2})^{(2)} = 0.$$

Because

$$\begin{Bmatrix} 4 & 4 & 2 \\ 5/2 & 5/2 & 5/2 \end{Bmatrix} = -\frac{\sqrt{33}}{126} \neq 0,$$

the latter is equivalent to $[A^{(4)} \otimes A^{(4)}]^{(2)} = 0$.

Define $G_{ab} = \frac{1}{2}\operatorname{tr}[(PJ_aP)(PJ_bP)]$. The $I/A^{(4)}$ cross terms have no traceless rank-two part by tensor-rank orthogonality. The rank-two part of the $A^{(4)}/A^{(4)}$ term is proportional to $[A^{(4)} \otimes A^{(4)}]^{(2)}$, which vanishes. Hence $G = gI_3$. For a rank-L tensor operator,

$$\sum_a J_a X^{(L)} J_a = \left[j(j+1) - \frac{L(L+1)}{2}\right] X^{(L)}.$$

Using $j = 5/2$, $L = 4$, $\|A^{(4)}\|^2 = 4/3$, and $\operatorname{tr} J^2 = 105/2$ gives

$$3g = \frac{1}{2}\left[\frac{1}{9}\operatorname{tr} J^2 - \frac{5}{4}\|A^{(4)}\|^2\right] = \frac{25}{12}, \qquad G = \frac{25}{36} I_3.$$

Since P is time-even and J_a is time-odd, $\operatorname{tr}(PJ_aP) = \operatorname{tr}(PJ_a) = 0$. Thus on the Kramers plane write $PJ_aP = v_a \cdot \boldsymbol{\sigma}$. If V has rows v_a, then $VV^T = 25I/36$, hence $V^TV = 25I/36$. Every pure codeword has a unit Bloch vector r and $s = Vr$, so $|s|^2 = 25/36$.

The state (44) has

$$\langle J_z \rangle = \frac{5}{6}, \qquad \langle J_z^2 \rangle = \langle J_x^2 \rangle = \langle J_y^2 \rangle = \frac{35}{12}, \qquad \langle K_{ab} \rangle = 0 \quad (a \neq b).$$

Indeed, its occupied J_z levels differ by four, so operators quadratic in J have no cross terms, while $J^2 = 35I/4$ fixes the two equal transverse moments. Thus it has $Q = 0$ and, after rotation, the same moments as the original state. □

Theorem 3.3 (Global C_2 reduction)**.** *For every $0 < S_1 \leq 9/5$, some global fixed-S_1 minimizer of S_2 has Dicke parity, equivalently a collective C_2 symmetry about its mean-spin axis.*

Proof. For $0 < S_1 < 9/5$, the KKT argument proves alignment. A nonaxial even–odd crossing cannot contain an aligned genuine parity mixture. The axial mixed branches are nonoptimal except at the scalar point, where Lemma 3.2 supplies a parity state with identical moments. Hence a parity minimizer exists. At $S_1 = 9/5$, (35) gives $S_2 \geq 2S_1 = 18/5$, and $D_5^{(1)}$ attains equality. □

Theorem 3.3 and Proposition 2.2 prove the first two lower-boundary pieces for unrestricted states.

4 Remaining edges, filling, and nonconvexity

The linear lower edge

The first identity in (35) gives the universal inequality $S_2 \geq 2S_1$. Equality is equivalent to $p_1 = 0$. By Lemma 3.1, one Majorana point then has multiplicity at least four, and (37) gives $S_1 \geq 9/5$. Conversely, the family (11) satisfies

$$S_1 = 5 - \frac{16}{5}\sin^4\theta, \qquad S_2 = 2S_1,$$

and covers precisely $9/5 \leq S_1 \leq 5$. This proves the third lower piece. Moreover, after rotating the fourfold Majorana point to the north pole, every equality state belongs to $\mathrm{span}\{D_5^{(0)}, D_5^{(1)}\}$. A collective z-rotation and a global phase put it in the form $d(\theta)$. Thus (11) gives all equality states up to collective rotations and global phase. The same multiplicity estimate also gives, for $r = 3, 4, 5$,

$$S_1 \geq \frac{1}{5}, \quad \frac{9}{5}, \quad 5,$$

respectively.

The upper and left edges

We prove $S_2 \leq 10$ without invoking a shadow inequality. Let C be a real traceless symmetric 3×3 matrix with $\mathrm{tr}\, C^2 = 1$. Equation (33), applied to C, implies that every eigenvalue β of $C : K$ obeys

$$\beta^3 - 14\beta = \frac{80}{3}\,\mathrm{tr}(C^3).$$

If $c_1 + c_2 + c_3 = 0$ and $\sum c_i^2 = 1$, then

$$|\,\mathrm{tr}(C^3)| = |3c_1c_2c_3| \leq \frac{1}{\sqrt{6}},$$

with equality when two eigenvalues coincide. Since $z^3 - 14z$ is increasing for $z \geq \sqrt{50/3}$ and

$$\left(\sqrt{\frac{50}{3}}\right)^3 - 14\sqrt{\frac{50}{3}} = \frac{80}{3\sqrt{6}},$$

we obtain

$$\lambda_{\max}(C : K) \leq \sqrt{\frac{50}{3}}.$$

For $Q \neq 0$, choose $C = Q/\sqrt{\mathcal{Q}}$. Since Q is traceless,

$$\mathcal{Q} = Q : M = \langle Q : K\rangle \leq \sqrt{\mathcal{Q}}\sqrt{\frac{50}{3}}.$$

Thus $\mathcal{Q} \leq 50/3$, also trivially when $Q = 0$, and (1) gives $S_2 \leq 10$. The GHZ family (12) has

$$S_1 = 5\cos^2\varphi, \qquad S_2 = 10,$$

so it realizes the whole upper edge.

The state ψ_0 in Section 1 gives $(S_1, S_2) = (0, 18/5)$, while the equal-amplitude GHZ state gives $(0, 10)$. To prove the lower endpoint at $S_1 = 0$, suppose that $s = 0$ and perturb ψ to the normalized vector proportional to $\psi + \varepsilon J_z\psi$. Then

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\langle J_z\rangle\Big|_{\varepsilon=0} = 2\langle J_z^2\rangle > 0$$

for half-integer spin. The perturbed states have $S_1 > 0$ for all sufficiently small nonzero ε. Continuity of S_2 and of the positive-S_1 lower boundary, whose limit is $18/5$, rules out $S_2 < 18/5$ at $S_1 = 0$.

There are no holes

The collective $SU(2)$ action on $\mathbb{CP}^5$ is Hamiltonian, with moment map

$$\mathbf{\Phi}([\psi]) = \langle\psi|\boldsymbol{J}|\psi\rangle = s.$$

Let π_+ send each point of $\mathfrak{su}(2)^*$ to the unique representative of its coadjoint orbit in the closed positive Weyl chamber, and put $\Psi = \pi_+ \circ \mathbf{\Phi}$. Kirwan's connectedness theorem states that every fiber of Ψ is connected [1]. For $SU(2)$, after identifying the positive Weyl chamber with $[0, \infty)$,

$$\Psi([\psi]) = |s|.$$

Hence the fixed-$S_1 = x$ set is the connected compact fiber

$$\Psi^{-1}\left(\sqrt{\frac{5x}{4}}\right) = \mathbf{\Phi}^{-1}\left(\left\{s : |s| = \sqrt{\frac{5x}{4}}\right\}\right).$$

The function S_2 is continuous and rotationally invariant, so its image on this fiber is a compact interval. Its minimum is $b(x)$, and the GHZ family supplies its maximum 10. Therefore the image is exactly $[b(x), 10]$. This proves (7), including every intermediate point of the left edge, and completes the proof of Theorem 1.2.

Nonconvexity

Proposition 4.1. *The set $\mathcal{S}_5^{\mathrm{sym}}$ is not convex.*

Proof. On the first lower-boundary arc, $x(p) = u(p)^2/5$ and $b(x(p)) = \Phi(p, \rho(p))$. Exact differentiation gives

$$\frac{\mathrm{d}b}{\mathrm{d}x} = \frac{2(p - \frac{3\sqrt{5}}{5})(p + \sqrt{5})}{p^2 + \frac{2\sqrt{5}}{5}p + \frac{9}{5}},$$

$$\frac{\mathrm{d}}{\mathrm{d}p}\left(\frac{\mathrm{d}b}{\mathrm{d}x}\right) = \frac{96(p + \sqrt{5}/5)}{5(p^2 + \frac{2\sqrt{5}}{5}p + \frac{9}{5})^2} > 0.$$

On $p_c < p < p_0$, one has $u(p) > 0$ and $u'(p) < 0$ by (22); hence $x'(p) = 2u(p)u'(p)/5 < 0$. Therefore

$$b''(x) < 0 \qquad (0 < x < x_c).$$

Choose $0 < x_1 < x_2 < x_c$. Strict concavity gives

$$b\left(\frac{x_1 + x_2}{2}\right) > \frac{b(x_1) + b(x_2)}{2}.$$

The points $P_i = (x_i, b(x_i))$ are attainable, but their midpoint lies strictly below the lower boundary. Thus $(P_1 + P_2)/2 \notin \mathcal{S}_5^{\mathrm{sym}}$. □

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